

## 4.2.3 Structure and bonding of carbon

### 4.2.3.1 Diamond

| Content                                                                                                                                                                                                | Key opportunities for skills development                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| In diamond, each carbon atom forms four covalent bonds with other carbon atoms in a giant covalent structure, so diamond is very hard, has a very high melting point and does not conduct electricity. | MS 5b<br>Visualise and represent 2D and 3D forms including two-dimensional representations of 3D objects. |
| Students should be able to explain the properties of diamond in terms of its structure and bonding.                                                                                                    | WS 1.2                                                                                                    |

### 4.2.3.2 Graphite

| Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Key opportunities for skills development |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| <p>In graphite, each carbon atom forms three covalent bonds with three other carbon atoms, forming layers of hexagonal rings which have no covalent bonds between the layers.</p> <p>In graphite, one electron from each carbon atom is delocalised.</p> <p>Students should be able to explain the properties of graphite in terms of its structure and bonding.</p> <p>Students should know that graphite is similar to metals in that it has delocalised electrons.</p> | WS 1.2                                   |

#### 4.2.3.3 Graphene and fullerenes

| Content                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Key opportunities for skills development                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <p>Graphene is a single layer of graphite and has properties that make it useful in electronics and composites.</p> <p>Students should be able to explain the properties of graphene in terms of its structure and bonding.</p> <p>Fullerenes are molecules of carbon atoms with hollow shapes. The structure of fullerenes is based on hexagonal rings of carbon atoms but they may also contain rings with five or seven carbon atoms. The first fullerene to be discovered was Buckminsterfullerene (<math>C_{60}</math>) which has a spherical shape.</p> <p>Carbon nanotubes are cylindrical fullerenes with very high length to diameter ratios. Their properties make them useful for nanotechnology, electronics and materials.</p> <p>Students should be able to:</p> <ul style="list-style-type: none"><li>• recognise graphene and fullerenes from diagrams and descriptions of their bonding and structure</li><li>• give examples of the uses of fullerenes, including carbon nanotubes.</li></ul> | <p>WS 1.2, 1.4</p> <p>MS 5b</p> <p>Visualise and represent 2D and 3D forms including two-dimensional representations of 3D objects.</p> |